Transmission from a samples-saturated teacher

Exact canon transfer from finitely many reconstructions, an Ω(1/n) barrier for every raw-data learner, and where the hoped-for advantage stops

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0. Results at a glance

Setting. A neural decoder $D_\theta : K \to \mathbb{R}^d$ with canon $M_\theta = D_\theta(K)$, an exact projection oracle π_θ, reconstruction risk $R_P(\theta) = \frac{1}{2}\mathbb{E}_P\|X - \pi_\theta(X)\|^2$. A teacher θ_* has been trained on P and frozen; $Q = (\pi_{\theta_*})_\# P$. One student learns from raw samples $X_i \sim P$, the other from reconstructions $Y_i = \pi_{\theta_*}(X_i)$. Same architecture, latent domain, capacity class. Following the suggestion to leave the online setting, the main results are for **batch training to convergence** on n samples (empirical-risk minimisation); the online version is in §9.

The mechanism is the batch counterpart of the vanishing-gradient-noise observation. Every reconstruction lies exactly on a canon that the architecture can represent, so all n per-sample losses have a **common exact zero** at the teacher. The empirical minimiser therefore does not move with the sample. Raw per-sample losses have no common zero ($\Sigma \neq 0$), so the raw empirical minimiser fluctuates by $\Theta(n^{-1/2})$ and overfits.

Proved.

#	Statement	Type
Thm 1	**Exact transmission.** If the teacher's canon is *regular* (embedded, interior latent support), *Q-rigid* (its only first-order deformations that keep it through the reconstructions are parameter symmetries), and the zero set of R_Q is the symmetry orbit, then there are $C, c > 0$ such that, for every n, with probability $\geq 1 - Ce^{-cn}$, **every** student that fits the n reconstructions exactly has exactly the teacher's canon — the *whole* canon, including parts of the repertoire that the data never exercise. Consequences: all four targets at once — zero excess real-data risk, exact imitation $\pi_{\hat\theta} = \pi_{\theta_*}$ on fresh real inputs, $R_Q = 0$, and canon recovery — plus zero generalisation gap. No optimality of the teacher is used, and n_0 does not depend on the noise level of P.	statistical (ERM), certifiable
Lem 6.2	If the teacher is the **unique global minimiser modulo symmetries**, the orbit condition holds automatically, and the global identifiability margin of the reconstruction problem is controlled by that of the raw problem.	structural
Prop 6.1	**Rigidity from sharpness:** $\lambda_{\min}(H_Q\|_N) \;\geq\; \lambda_{\min}(H_P\|_N) - C_2\sqrt{2R_*}$. A strict teacher whose sharpness dominates its residual–curvature coupling is Q-rigid.	local geometry
Thm 2	**No raw learner beats $1/n$.** If the teacher is a nondegenerate unique global minimiser and $R_* > 0$ (with an output bias), then *every* procedure using n raw samples has worst-case expected excess risk $\geq a/(64n)$ over two nearby worlds. The raw ERM overfits by $\approx \operatorname{tr}(H^+\Sigma)/n$.	minimax lower bound

#	Statement	Type
Thm 3	**Dichotomy.** Combining the two: $n_{\text{rec}}(\varepsilon) = O(\log(1/\varepsilon))$, and exact transmission at a finite n_0, versus $n_{\text{raw}}(\varepsilon) \geq a/(64\varepsilon)$.	statistical separation
Thm 4	**Imperfect teachers.** The reconstruction student reproduces the teacher's canon, and hence exactly the teacher's real-data risk, for every $n \geq n_0$. It never improves on the teacher. The advantage window is $n_0 \leq n \lesssim N_{\text{eff}} = \operatorname{tr}(H^{+}\Sigma)/(2\Delta_{\text{teacher}})$; for a teacher trained on N raw samples, $N_{\text{eff}} \approx N$. n_0 **reconstructions transmit what** N **raw samples taught the teacher.**	statistical
Thm 5	Online: constant-step SGD on reconstructions converges linearly inside the rigidity basin; Thm 2 bounds raw SGD with any schedule.	algorithmic (local) vs statistical

Obstructions (each made precise below):

- **O1 — memorisation threshold (Prop 7).** At sample sizes where a raw student can exactly interpolate its sample near the teacher, the set of exact fits to reconstructions is itself a manifold of dimension $p + nk - nd$ larger than the orbit. So the strongest reading of "raw training fully overfits while reconstruction training fully doesn't" fails for *all* minimisers there, and needs an implicit-bias hypothesis. Exactness needs $n(d-k) \geq p - \dim(\text{orbit})$. What survives in that regime is a uniform benignness bound for exact fits to reconstructions (Prop 2.3).
- **O2 — imperfect teachers are never surpassed (Thm 4).** This explains the digits-pilot trade-off.
- **O3 — basin access is not implied (Prop 8).** If P has a symmetry that the capacity-limited canon must break, every inequivalent optimal canon is (for small residuals) a spurious local minimum of the reconstruction objective with *constant* excess risk, where raw training is fine. The training-loss certificate (Cor 3.2) detects this failure; raw training has no such certificate.
- **O4 — conditioning and basin access (numerics).** In a tanh decoder Q-rigidity holds, but it is badly conditioned ($\lambda_{\min}/\lambda_{\max} \approx 5 \times 10^{-11}$ on the normal space). From independent random initialisations, gradient training on reconstructions did not reach an exact fit within our budget. It stalled in spurious basins, which its training loss flagged. Without a capacity bound, a wider decoder approaches zero loss only along valleys in which the weights grow.

Numerics (§10; a tanh decoder on a circle latent in $\mathbb{R}^6$, $p = 78$, samples-saturated teacher).

- **Every exact fit, from the predicted threshold on.** Every exact fit to $n \geq 16$ reconstructions has real-data excess at the evaluation's resolution ($\leq 3 \times 10^{-7}$). Exact fits to $n \leq 12$ do not, matching the dimension count $(p-1)/(d-k) = 15.4$ of Prop 7.
- **Raw students** follow the predicted $\frac{1}{2}\operatorname{tr}(H^{+}\Sigma)/n$ within a factor ≈ 2, and overfit with a test–train gap of about twice their excess.
- **In-basin students.** With students started in the teacher's basin and selected by their own training loss, the separation is 3–6 orders of magnitude for $24 \leq n \leq 1024$.
- **Independent random inits.** Reconstruction students still win at every $n \leq 1024$ (by $14\times$ down to $1.6\times$), but they plateau at their optimisation floor, so the advantage there is a finite window.
- **Imperfect teachers** are reproduced exactly, with the predicted window.
- **Symmetry-breaking traps** appear exactly as predicted.

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1. Setting

1.1 Objects. K is a compact connected C^3 manifold (the latent domain). The capacity class $\Theta \subset \mathbb{R}^p$ is compact, and $D : \Theta' \times K \to \mathbb{R}^d$ is C^3 on an open $\Theta' \supset \Theta$. Put $M_\theta = D_\theta(K)$ (compact), $\ell_\theta(x) = \frac{1}{2}\operatorname{dist}(x, M_\theta)^2$, and $R_\mu(\theta) = \mathbb{E}_\mu\, \ell_\theta$ for any law μ. P has compact support. The teacher is $\theta_* \in \operatorname{int}\Theta$, $M_* := M_{\theta_*}$, π_* is a measurable nearest-point selection onto M_*, $Q = (\pi_*)_\# P$, and $R_* := R_P(\theta_*)$. Teacher optimality is assumed only where stated.

1.2 Symmetries. G is a compact group acting on $\mathbb{R}^p$ by orthogonal maps that preserve Θ, with diffeomorphisms ρ_g of K such that $D_{g\theta} = D_\theta \circ \rho_g$. Examples: hidden-unit permutations; sign flips for odd activations; latent isometries realised by the first layer. Then $M_{g\theta} = M_\theta$ and $\ell_{g\theta} = \ell_\theta$. The *orbit* is $G\theta_*$.

1.3 The zero set. $Z := \{\theta \in \Theta :\ R_Q(\theta) = 0\} = \{\theta \in \Theta :\ \operatorname{supp} Q \subseteq M_\theta\}$ (closed). Always $G\theta_* \subseteq Z$.

1.4 Learners (batch). Draw i.i.d. $X_1, \dots, X_n \sim P$ and put $Y_i = \pi_*(X_i)$. The raw student minimises $\hat R_n^P(\theta) = \frac{1}{n}\sum \ell_\theta(X_i)$ and the reconstruction student minimises $\hat R_n^Q(\theta) = \frac{1}{n}\sum \ell_\theta(Y_i)$, both over Θ. "Training fully" is idealised as reaching a global minimiser. Cor 3.2 turns this into a verifiable condition, and §9 discusses algorithms.

1.5 Local objects at a canon point. At (θ, z) write $J = \partial_\theta D_\theta(z) \in \mathbb{R}^{d\times p}$ and $A = \partial_z D_\theta(z) \in \mathbb{R}^{d\times k}$. $P^\perp$ is the orthogonal projector onto $(\operatorname{range} A)^\perp$, the normal space of the canon. For $y = D_{\theta_0}(z_y) \in M_{\theta_0}$ define the **normal Gauss–Newton matrix**

$$G(y;\theta_0) = J^\top P^\perp J \ \succeq 0, \qquad H_Q(\theta_0) = \mathbb{E}_Q\, G(Y;\theta_0).$$

Also define $H_P = \nabla^2 R_P(\theta_*)$ (when it exists) and $\Sigma = \mathbb{E}_P[\nabla\ell_{\theta_*}(X)\nabla\ell_{\theta_*}(X)^\top]$. N_{θ_0} is the orthogonal complement of $T_{\theta_0}(G\theta_0)$. By G-invariance, $\nabla\ell_{\theta_*}(x) \in N_{\theta_*}$ for every x, so Σ lives on $N := N_{\theta_*}$.

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2. Universal facts (no regularity, any teacher)

Proposition 2.1 (two-sided transfer). For every θ and x,

$$\operatorname{dist}(x, M_\theta) \le \|x - \pi_*(x)\| + \operatorname{dist}(\pi_*(x), M_\theta),$$

and therefore

$$\sqrt{R_P(\theta)} \le \sqrt{R_*} + \sqrt{R_Q(\theta)}, \qquad \sqrt{R_Q(\theta)} \le \sqrt{R_*} + \sqrt{R_P(\theta)}.$$

Proof. The first line is the triangle inequality through the point of M_θ nearest to $\pi_*(x)$. Minkowski's inequality in $L^2(P)$ gives the second. For the third, use $\operatorname{dist}(\pi_*(x), M_\theta) \le \|\pi_*(x) - x\| + \operatorname{dist}(x, M_\theta)$. $\square$

The left inequality is the stated transfer inequality ($R_P - R_* \le R_Q + 2\sqrt{R_* R_Q}$). The right one shows that the two square-root risk landscapes are uniformly within $\sqrt{R_*}$ of each other; it is used in Lemma 6.2 and Prop 8.

Proposition 2.2 (interpolation; globally self-bounding gradients). Let $y \in \mathbb{R}^d$, $z^\star \in \arg\min_z \|y - D_\theta(z)\|$, and let $g = -J_\theta(z^\star)^\top (y - D_\theta(z^\star))$ be the envelope gradient used by SGD. If $\phi(\theta') = \frac{1}{2}\|y - D_{\theta'}(z^\star)\|^2$ has L-Lipschitz gradient on the segment $[\theta, \theta - g/L]$, then

$$\|g\|^2 \le 2L\, \ell_\theta(y).$$

In particular, for $y \in M_*$: $\ell_{\theta_*}(y) = 0$ and $g = 0$.

Proof. $\ell_{\theta'}(y) \le \phi(\theta') \le \phi(\theta) + \langle g, \theta' - \theta\rangle + \frac{L}{2}\|\theta' - \theta\|^2$. Take $\theta' = \theta - g/L$ and use $\ell \ge 0$. □

So reconstruction-gradient noise is multiplicative everywhere, not only near the teacher. For raw data, $\mathbb{E}\|g_{\theta_*}(X)\|^2 = \operatorname{tr}\Sigma$, and with a trainable output bias c one has $\nabla_c \ell_{\theta_*}(x) = -(x - \pi_*(x))$, so $\Sigma \neq 0$ whenever $R_* > 0$.

Proposition 2.3 (exact fits to reconstructions are uniformly benign). For *any* θ, of any architecture or capacity, with $Y_1, \dots, Y_n \in M_\theta$:

$$R_Q(\theta) \le q_n := \tfrac{1}{2}\,\mathbb{E}_{Y\sim Q} \min_{i\le n} \|Y - Y_i\|^2, \qquad R_P(\theta) \le \left(\sqrt{R_*} + \sqrt{q_n}\right)^2.$$

We have $q_n \to 0$ almost surely, and $q_n = O_P\big((\log n/n)^{2/k}\big)$ if Q has a density bounded below on a k-dimensional canon.

Proof. $\operatorname{dist}(Y, M_\theta) \le \min_i \|Y - Y_i\|$, then Prop 2.1. □

There is no analogue for exact fits to raw data. The corresponding quantity $\frac{1}{2}\mathbb{E}\min_i \|X - X_i\|^2$ scales with the intrinsic dimension of P (typically d), not that of the canon.

Proposition 2.4 (analytic decoders: no hallucination; exact imitation on Z). Assume K is a compact connected real-analytic manifold without boundary, each D_θ is real-analytic on K, D_* is an analytic embedding, and $\mathcal{H}^k(\operatorname{supp} Q) > 0$ (e.g. Q has a density on M_*). Then every $\theta \in Z$ satisfies

$$\operatorname{supp} Q \subseteq M_\theta \subseteq M_*.$$

Consequently, for every $\theta \in Z$: $\operatorname{dist}(X, M_\theta) = \operatorname{dist}(X, M_*)$ P-a.s., $R_P(\theta) = R_*$, and if nearest points on M_* are P-a.s. unique, $\pi_\theta(X) = \pi_*(X)$ P-a.s.

Proof. Let V be a tubular neighbourhood of M_* on which the normal-coordinate map $\nu(x) = x - \pi_*(x)$ is analytic, so $M_* = V \cap \{\nu = 0\}$. The set $E = D_\theta^{-1}(M_*)$ is closed in K and analytic in the open set $D_\theta^{-1}(V)$. If $\dim E < k$, then E is a countable union of analytic submanifolds of dimension $< k$, so $\mathcal{H}^k(D_\theta(E)) = 0$. That contradicts $D_\theta(E) = M_\theta \cap M_* \supseteq \operatorname{supp} Q$. Hence E contains an open set W. On the connected component $C \ni W$ of $D_\theta^{-1}(V)$, the analytic map $\nu \circ D_\theta$ vanishes on W, hence on C. Next, C is closed in K: a limit point w has $D_\theta(w) \in M_* \subset V$, so a connected neighbourhood of w inside $D_\theta^{-1}(V)$ meets C and is therefore contained in C. So $C = K$, i.e. $D_\theta(K) \subseteq M_*$. For the consequences, $\pi_*(X) \in \operatorname{supp} Q \subseteq M_\theta$ a.s. gives $\operatorname{dist}(X, M_\theta) \le \operatorname{dist}(X, M_*)$, and $M_\theta \subseteq M_*$ gives the reverse. A nearest point of M_θ is then a nearest point of M_*. □

So, for analytic decoders, reaching $R_Q = 0$ already *is* exact functional transmission of the teacher's reconstruction map (the user's target (b)). This holds for any teacher, optimal or not. The student's canon can be no larger than the teacher's.

3. Exact transmission from finitely many reconstructions

Assumptions.

- **(A1) Capacity.** Θ is compact, $D \in C^3$ near $\Theta \times K$, and $\theta_* \in \operatorname{int}\Theta$.
- **(A2) Regular canon.** D_* is an embedding (injective immersion), and the latent support $S_* := D_*^{-1}(\operatorname{supp} Q) \subset \operatorname{int} K$ (automatic if $\partial K = \emptyset$).
- **(A3) Orbit zero set.** $Z = G\theta_*$: within the capacity class, the canon on $\operatorname{supp} Q$ determines the parameters up to symmetry. *By Lemma 6.2 this is automatic when the teacher is the unique global minimiser of R_P modulo G.*
- **(A4) Q-rigidity.** $\mu := \lambda_{\min}\big(H_Q(\theta_*)|_N\big) > 0$. Equivalently $\ker H_Q(\theta_*) = T_{\theta_*}(G\theta_*)$: the only infinitesimal parameter deformations that move the canon tangentially along $\operatorname{supp} Q$ are symmetry directions. Prop 6.1 derives this from teacher sharpness.

Theorem 1 (exact transmission). Under (A1)–(A4) there are constants $C, c > 0$, depending on the architecture, Θ, M_* and Q but not on n, such that for every n, with probability at least $1 - Ce^{-cn}$,

$$\arg\min_{\Theta} \hat{R}_n^Q = \{\theta \in \Theta : \hat{R}_n^Q(\theta) = 0\} = G\theta_*.$$

On this event every reconstruction-trained student $\hat{\theta}$ satisfies $M_{\hat{\theta}} = M_*$. Hence $R_Q(\hat{\theta}) = \hat{R}_n^Q(\hat{\theta}) = 0$ (no generalisation gap), $R_P(\hat{\theta}) = R_*$, and $\pi_{\hat{\theta}} = \pi_*$ (up to the tie-breaking rule; P-a.s. if projections are a.s. unique). In particular $\mathbb{E}[R_P(\hat{\theta})] - R_* \le B_0 C e^{-cn}$, where $B_0 = \sup_{\Theta \times \operatorname{supp} P} \ell$.

The constants are explicit. Let $B_G = \sup_y \|G(y;\theta_*)\|$, let r_* and C_3 be as in Lemma 3.1, let $\gamma = \min\{R_Q(\theta) : \operatorname{dist}(\theta, G\theta_*) \ge r_*\}$, and let N_γ, m_γ be the number and minimal Q-mass of a cover of $\operatorname{supp} Q$ by balls of radius $\sqrt{\gamma}/2$ centred in $\operatorname{supp} Q$. Then one can take

$$C = (p - \dim G\theta_*) + N_\gamma, \qquad c = \min\Big\{\tfrac{\mu}{2B_G}\log\tfrac{e}{2},\ m_\gamma\Big\}.$$

Lemma 3.1 (uniform local structure). There are $r_1 > 0$ and $C_3 < \infty$ with the following properties.

- Every θ with $\operatorname{dist}(\theta, G\theta_*) \le r_1$ is uniquely $\theta_0 + v$ with $\theta_0 \in G\theta_*$, $v \in N_{\theta_0}$, $|v| = \operatorname{dist}(\theta, G\theta_*)$.
- For all such θ and all $y \in \operatorname{supp} Q$, the projection of y onto M_θ is unique and interior, depends C^2 on θ, and

$$\ell_{\theta_0+v}(y) \ \ge\ \tfrac{1}{2}\, v^\top G(y;\theta_0)\, v - C_3|v|^3.$$

Proof. The orbit of a compact Lie group is a compact smooth submanifold with positive reach, which gives the decomposition. By invariance ($\ell_{g\theta} = \ell_\theta$, $gN_{\theta_*} = N_{g\theta_*}$) it suffices to take $\theta_0 = \theta_*$. Let $y = D_*(z_y)$ with $z_y \in S_*$.

- *Minimisers stay near z_y.* Since D_* is an embedding of compact K, $\|D_*(z) - D_*(z')\| \ge \omega(|z - z'|)$ with $\omega > 0$ on $(0, \infty)$. If $|\theta - \theta_*| \le r$, every minimiser z of $F(z) = \tfrac{1}{2}\|y - D_\theta(z)\|^2$ has $\|D_\theta(z) - y\| \le Lr$, hence $|z - z_y| \le \omega^{-1}(2Lr)$.

- *Strong convexity there.* Near z_y, $\partial_z^2 F = A^\top A - \langle y - D_\theta(z), \partial_z^2 D\rangle \succeq \frac{\alpha}{2} I$ for r small, where $\alpha = \min_K \lambda_{\min}(A_*^\top A_*) > 0$. So the minimiser is unique and interior and, by the implicit function theorem, C^2 in θ, uniformly over the compact S_*.
- *Second-order expansion.* $\ell_\theta(y)$ is then C^3, and at θ_* it has value 0, gradient 0 (zero residual) and Hessian $G(y;\theta_*)$ (Appendix A). Taylor's theorem with a uniform bound on third derivatives gives the inequality. $\square$

Proof of Theorem 1.

1. **Concentration.** $\hat G_n := \frac{1}{n}\sum_i G(Y_i;\theta_*)$ is an average of i.i.d. PSD matrices with $\|\cdot\| \le B_G$ and $\lambda_{\min}(\mathbb{E}\,\cdot\,|_N) = \mu$. The matrix Chernoff inequality (Tropp 2012, Thm 1.1, with $\delta = \frac12$) applied to the compressions to N gives

$$\Pr\left[\lambda_{\min}(\hat G_n|_N) \le \mu/2\right] \le (\dim N)\,(2/e)^{n\mu/(2B_G)}.$$

   Since $\hat G_n(g\theta_*) = g\hat G_n(\theta_*)g^\top$, on the complement $\lambda_{\min}(\hat G_n(\theta_0)|_{N_{\theta_0}}) > \mu/2$ simultaneously for all $\theta_0 \in G\theta_*$.

2. **Local exactness.** Put $r_* := \min\{r_1, \mu/(8C_3)\}$. On that event, averaging Lemma 3.1 over the sample gives, for $\theta = \theta_0 + v$ with $0 < |v| \le r_*$,

$$\hat R_n^Q(\theta) \;\ge\; \tfrac{\mu}{4}|v|^2 - C_3|v|^3 \;\ge\; \tfrac{\mu}{8}|v|^2 > 0.$$

   So exact fits within distance r_* of the orbit lie on the orbit.

3. **Global localisation.** R_Q is continuous on the compact Θ, since $|\mathrm{dist}(y, M_\theta) - \mathrm{dist}(y, M_{\theta'})| \le \sup_z \|D_\theta(z) - D_{\theta'}(z)\|$. It vanishes exactly on $Z = G\theta_*$, so $\gamma > 0$.
   - If $\hat R_n^Q(\theta) = 0$, then $Y_i \in M_\theta$ for all i, so by Prop 2.3 $R_Q(\theta) \le q_n$.
   - If every ball of the $\sqrt{\gamma}/2$-cover contains a sample point, then $q_n \le \gamma/2 < \gamma$. This fails with probability at most $N_\gamma(1 - m_\gamma)^n$.
   - Otherwise $R_Q(\theta) < \gamma$ forces $\mathrm{dist}(\theta, G\theta_*) < r_*$, and step 2 then forces $\theta \in G\theta_*$.

4. **Union bound.** Combine the two failure events. Since $\hat R_n^Q \ge 0$ vanishes on $G\theta_*$, the minimisers are exactly the zeros. The consequences follow from $M_{g\theta_*} = M_*$. $\square$

Remarks.

1. **No noise in n_0.** Teacher optimality was not used. Nor does the noise level of P enter. If y is a nearest point of a closed set to x, it is the *unique* nearest point to every point of the open segment (y, x). Hence for the family $P_\lambda := \mathrm{law}\left(\pi_*(X) + \lambda(X - \pi_*(X))\right)$, $0 < \lambda \le 1$, the reconstruction law is the same Q for every λ. Also $\mathbb{E}_{P_\lambda}\nabla\ell_{\theta_*} = \lambda\,\mathbb{E}_P\nabla\ell_{\theta_*}$, so a stationary teacher stays stationary, and $\Sigma_\lambda = \lambda^2\Sigma$. So n_0 is literally independent of λ, while the raw lower-bound constant of §4 scales like λ^2.
2. **Compactness is essential.** γ is a *global* margin. With unbounded weights there can be near-zero valleys at infinity, i.e. canons that approximate M_* arbitrarily well with diverging parameters, so that $\gamma = 0$. We observe these numerically (§10, O4). Lemma 6.2 bounds γ below by the raw margin.
3. **Boundaries.** If K has a boundary and P puts mass "beyond" it, Q has singular mass on $D_*(\partial K)$, violating (A2). If instead $S_* \subset \mathrm{int}\,K$ but K is a box, Z contains latent sub-box reparametrisations ($z \mapsto D_*(\Lambda z + m)$ with $\Lambda K + m \supseteq S_*$), whose boundary points are not regular. There exactness

can fail, but only through "slivers" of missing coverage. The cost is (missing mass)×(width)^2, e.g. $O((\log n/n)^3)$ for $k=1$, still $o(1/n)$. A boundaryless latent space (circle, torus, sphere) avoids this.

4. **General zero sets.** Without (A3), the same proof works if Z is a compact C^2 manifold along which (A2) and (A4) hold, using a net of Z in step 1. The conclusion is then $\arg\min \hat{R}_n^Q = Z$, and Props 2.1/2.4 give $R_P(\hat{\theta}) \le R_*$, with equality and exact imitation for analytic decoders.

Corollary 3.2 (a training-loss certificate). Assume in addition that θ_* is a stationary point of R_P, and let L_P be a semiconcavity constant of R_P near $G\theta_*$. (R_P is a minimum of uniformly $C^{1,1}$ functions, so such an L_P exists.) There are $\tau_0 > 0$ and $n_1 < \infty$ such that for $n \ge n_1$, with probability $\ge 1 - C'e^{-c'n}$, **every** $\theta \in \Theta$ with $\hat{R}_n^Q(\theta) \le \tau \le \tau_0$ satisfies

$$\operatorname{dist}(\theta, G\theta_*) \le \sqrt{8\tau/\mu}, \qquad R_P(\theta) - R_* \le \frac{4L_P}{\mu}\,\tau.$$

Proof. The losses are uniformly Lipschitz in θ, so a net plus Hoeffding's inequality give $\sup_\Theta |\hat{R}_n^Q - R_Q| \le \varepsilon_n = O(\sqrt{p \log n/n})$ with high probability.

- If $\tau \le \gamma/2$ and $\varepsilon_n < \gamma/2$, then $R_Q(\theta) < \gamma$, so θ is within r_* of the orbit.
- Step 2 then gives $\tau \ge \frac{\mu}{8}\operatorname{dist}^2$.
- Finally, semiconcavity together with $\nabla R_P = 0$ on the orbit gives $R_P(\theta) \le R_* + \frac{L_P}{2}\operatorname{dist}^2$. □

So, for reconstructions, the training loss *certifies* transmission: the student can tell from its own data that it has succeeded. A raw student's training loss certifies nothing comparable, since the raw ERM's training loss sits *below* R_*.

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4. No raw learner beats $1/n$

Assumptions.

- **(B1) Saturated, unique teacher.** $\arg\min_\Theta R_P = G\theta_*$.
- **(B2) Tubular regularity.** For θ near $G\theta_*$, projections of P-almost every x onto M_θ are unique, interior and nondegenerate, e.g. $\operatorname{supp} P$ lies in the tube of radius $(1-\eta)\operatorname{reach}(M_*)$ and $D_*^{-1}(\pi_*(\operatorname{supp} P)) \subset \operatorname{int} K$. Then R_P and $\theta \mapsto \mathbb{E}_P[g\,\ell_\theta]$ (bounded g) are C^3 near $G\theta_*$.
- **(B3) Sharpness.** $\mu_P := \lambda_{\min}(H_P|_N) > 0$.
- **(B4) Noise.** $\Sigma \ne 0$. This holds as soon as $R_* > 0$ and there is a trainable output bias (Prop 2.2).

Theorem 2. Assume (A1) and (B1)–(B4), and let $a := \lambda_{\max}\big(\Sigma^{1/2} H_P^+ \Sigma^{1/2}\big) > 0$. There is n_1 such that for every $n \ge n_1$ there is a probability measure P_n with $dP_n/dP = 1 + t_n g$, where g is bounded and $t_n = O(n^{-1/2})$, with the following property. For **every** estimator $\hat{\theta}_n = \hat{\theta}_n(X_1, \dots, X_n)$, possibly randomised,

$$\max_{W \in \{P, P_n\}} \mathbb{E}_W\Big[R_W(\hat{\theta}_n) - \min_\Theta R_W\Big] \;\ge\; \frac{a}{64\,n}.$$

Proof. Fix $u \in N$ with $\Sigma u \ne 0$ and put $g(x) = u^\top \nabla \ell_{\theta_*}(x)$. It is bounded, and has mean zero *because the teacher is stationary*. Its second moment is $v_u := \mathbb{E}g^2 = u^\top \Sigma u$. Let $P_t = (1+tg)P$ and write $\mathcal{E}_t(\theta) = R_{P_t}(\theta) - \min R_{P_t} \ge 0$.

Separation. Put $\Delta = R_P - R_*$ and $\phi(\theta) = \mathbb{E}_P[g\,(\ell_\theta - \ell_{\theta_*})]$, so $R_{P_t} = R_* + t\,\mathbb{E}_P[g\ell_{\theta_*}] + \Delta + t\phi$. Let $m_s := -\inf_\theta[\Delta + s\phi] \geq 0$. Then, exactly,

$$\mathcal{E}_0(\theta) + \mathcal{E}_t(\theta) = 2\Delta(\theta) + t\phi(\theta) + m_t \;\geq\; m_t - 2m_{t/2}.$$

- ϕ is G-invariant with $\nabla\phi(\theta_*) = \Sigma u =: w \in N$. Put $a_u := w^\top H_P^+ w$.
- *Lower bound on m_t.* Plugging $\theta = \theta_* - tH_P^+ w$ gives $m_t \geq \frac{t^2}{2}a_u - O(t^3)$.
- *Upper bound on $m_{t/2}$.* Away from the orbit (distance $\geq \rho$), (B1) and compactness give $\Delta \geq \kappa_\rho > 0$, which beats $|t\phi| \leq tB_0\|g\|_\infty$ for small t. Near the orbit, invariance reduces to $\theta = \theta_* + \delta$ with $\delta \in N$. Then $\Delta + \frac{t}{2}\phi \geq \frac{1}{2}\delta^\top H_P \delta + \frac{t}{2}w^\top\delta - O(|\delta|^3 + t|\delta|^2)$, whose infimum is $\geq -\frac{t^2}{8}a_u(1 + O(\rho) + O(t))$.
- Hence, for ρ and t small, $m_t - 2m_{t/2} \geq t^2 a_u/8$.

Indistinguishability. For $|tg| \leq \frac{1}{2}$, $-\log(1+s) \leq -s + s^2$ gives $\mathrm{KL}(P^{\otimes n}\|P_t^{\otimes n}) \leq nt^2 v_u$. Take $t_n = (2nv_u)^{-1/2}$; Pinsker then gives $\mathrm{TV} \leq \frac{1}{2}$.

Le Cam. For nonnegative losses, $\mathbb{E}_0\mathcal{E}_0(\hat\theta) + \mathbb{E}_t\mathcal{E}_t(\hat\theta) \geq (1 - \mathrm{TV})\inf_\theta(\mathcal{E}_0 + \mathcal{E}_t)$. So the maximum is $\geq \frac{1}{4}\cdot\frac{t_n^2 a_u}{8} = \frac{a_u}{64\,n\,v_u}$. Maximise a_u/v_u over u to get a. □

Remarks.

1. **Sharp constant.** The local asymptotic minimax constant (Hájek–Le Cam) is $\frac{1}{2}\,\mathrm{tr}(H_P^+\Sigma)$, and the raw ERM attains it. Under the usual M-estimation conditions, $n(R_P(\hat\theta) - R_*) \rightsquigarrow \frac{1}{2}Z^\top H_P Z$ and $n(\hat R_n^P(\hat\theta) - \hat R_n^P(\theta_*)) \rightsquigarrow -\frac{1}{2}Z^\top H_P Z$, with $Z \sim \mathcal{N}(0, H_P^+\Sigma H_P^+)$. So the raw student's training risk sits $\approx \frac{1}{2}\,\mathrm{tr}(H^+\Sigma)/n$ *below* R_* and its test risk the same amount *above*. That is overfitting by $\approx \mathrm{tr}(H^+\Sigma)/n$ (Takeuchi's optimism).
2. **Applies to every raw procedure.** The bound holds for raw SGD with any schedule, ensembles, early stopping, and so on.

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5. The dichotomy

Theorem 3. Assume (A1), (A2), (A4) and (B1)–(B4); by Lemma 6.2, (A3) follows. For the worlds $\{P, P_n\}$ of Theorem 2, each with its own saturated teacher $\theta_*(W)$ and reconstructions $Y_i = \pi_{\theta_*(W)}(X_i)$:

1. In both worlds, every exact fit to n reconstructions has zero excess risk, with probability $\geq 1 - Ce^{-cn}$, where C, c are uniform in n.
2. Every raw-data procedure has $\max_W \mathbb{E}_W[\mathrm{excess}] \geq a/(64n)$.

Hence the sample complexity of ε-excess satisfies

$$n_{\mathrm{rec}}(\varepsilon) \leq c^{-1}\log(CB_0/\varepsilon), \qquad n_{\mathrm{rec}}^{\mathrm{exact}}(\delta) \leq c^{-1}\log(C/\delta),$$

$$n_{\mathrm{raw}}(\varepsilon) \;\geq\; a/(64\varepsilon).$$

Proof. (ii) is Theorem 2. For (i), the hypotheses are open conditions: $P_n \to P$ in total variation with a common support, $R_{P_n} \to R_P$ in C^3 near the orbit, so $\theta_*(P_n) \to \theta_*$, and μ, B_G, C_3, r_1, the covering data and the raw margin are continuous. So Theorem 1 applies to each P_n with uniform constants. □

This is a **statistical** separation. On the reconstruction side the procedure is a specific one, “fit the reconstructions exactly”, which Cor 3.2 makes verifiable; on the raw side the bound is against all procedures. The algorithmic side is in §9.

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6. Where rigidity and the orbit condition come from

Proposition 6.1 (curvature transfer). Assume (B2). Let

- b_J, b_A bound $\|J\|, \|A\|$,
- b_2 bound the second derivatives $\partial^2_{\theta\theta} D, \partial^2_{\theta z} D, \partial^2_{zz} D$ (as bilinear maps),
- $\alpha = \min \lambda_{\min}(A^\top A)$,

and suppose residuals satisfy $\|x - \pi_*(x)\| b_2 \le \alpha/2$. Then

$$\|H_P - H_Q(\theta_*)\| \;\le\; C_2\, \mathbb{E}_P \|X - \pi_*(X)\| \;\le\; C_2 \sqrt{2R_*}, \qquad C_2 = b_2\Big(2 + \frac{2b_J^2 b_A^2}{\alpha^2} + \frac{4 b_J b_A}{\alpha}\Big).$$

Hence $\lambda_{\min}(H_Q|_N) \ge \mu_P - C_2\sqrt{2R_*}$. Conversely, $\mu_P \ge \lambda_{\min}(H_Q|_N) - C_2\sqrt{2R_*}$.

Proof. At x with residual $r = x - \pi_*(x) \perp \operatorname{range} A$, the Hessian of the profiled loss is the Schur complement of the joint Hessian of $f(\theta, z) = \frac12 \|x - D_\theta(z)\|^2$:

$$h(x) = J^\top J - \langle r, D_{\theta\theta}\rangle - (J^\top A - R_1)(A^\top A - R_2)^{-1}(A^\top J - R_1^\top),$$

$$R_1 = \langle r, D_{\theta z}\rangle, \qquad R_2 = \langle r, D_{zz}\rangle.$$

- At $r = 0$ this is $J^\top P^\perp J = G(\pi_*(x); \theta_*)$.
- Expanding the difference with $\|(A^\top A - R_2)^{-1} - (A^\top A)^{-1}\| \le 2\|r\| b_2/\alpha^2$ gives $\|h(x) - G(\pi_*(x))\| \le C_2\|r\|$.
- Average over P: $H_P = \mathbb{E}_P h$, while $H_Q = \mathbb{E}_P G(\pi_*(X))$ because $Q = (\pi_*)_\# P$. □

Reading. H_Q measures first-order motion of the canon normal to itself on $\operatorname{supp} Q$. It is pure geometry: architecture plus teacher canon plus where the data project. H_P is the same matrix plus a residual-times-curvature coupling. So **Q-rigidity is the teacher’s sharpness, up to an error proportional to its residual scale**. In particular (A4) is implied by a property measurable at the trained teacher: strict minimality modulo symmetries with sharpness $\mu_P > C_2\sqrt{2R_*}$. It also gives the local linear transfer $R_P(\theta) - R_* \le \frac{2L_P}{\mu} R_Q(\theta)$ near the orbit (Cor 3.2), so the stated “additional regularity” is exactly (A4).

Lemma 6.2 (unique optimality gives the orbit condition, and controls the margin). Under (B1), $Z = G\theta_*$, i.e. (A3) holds. Moreover, with $\kappa_r := \min\{R_P(\theta) - R_* : \operatorname{dist}(\theta, G\theta_*) \ge r\} > 0$,

$$\gamma(r) := \min\{R_Q(\theta) : \operatorname{dist}(\theta, G\theta_*) \ge r\} \;\ge\; \big(\sqrt{R_* + \kappa_r} - \sqrt{R_*}\big)^2.$$

Proof. If $R_Q(\theta) = 0$, Prop 2.1 gives $R_P(\theta) \le R_* = \min R_P$, so $\theta \in G\theta_*$. For the bound, $\sqrt{R_Q} \ge \sqrt{R_P} - \sqrt{R_*}$ (Prop 2.1). □

So the global part of Theorem 1 costs nothing extra for a saturated, uniquely optimal teacher. The reconstruction problem is globally identifiable whenever the raw problem is, with margin at least $\kappa_r^2/(4(R_* + \kappa_r))$. What

(B1) does require is that P have no symmetry that the optimal canon must break. When it does, inequivalent optima exist and Prop 8 applies.

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7. Imperfect teachers: exact transmission of the teacher's risk

Theorem 4. Suppose θ_* satisfies (A1), (A2), (A4) and (B1)–(B2). There is a neighbourhood U of θ_* such that every teacher $\theta_T \in U$, *stationary or not*, satisfies the hypotheses of Theorem 1 with uniform constants. (For example, a teacher trained by raw ERM on N samples lies in U with probability $\to 1$ as $N \to \infty$.) Hence, for every n, with probability $\geq 1 - Ce^{-cn}$, every exact fit $\hat{\theta}$ to n reconstructions $\pi_{\theta_T}(X_i)$ has $M_{\hat{\theta}} = M_T$, and so

$$R_P(\hat{\theta}) = R_P(\theta_T) \quad \text{exactly}, \qquad \pi_{\hat{\theta}} = \pi_{\theta_T}.$$

Proof. (A2) and (A4) are open in θ_T. For (A3): under (B2), $\pi_{\theta_T} \to \pi_*$ uniformly on $\operatorname{supp} P$, so $R_{Q_T} \to R_Q$ uniformly on Θ. Since $\gamma > 0$, the zero set of R_{Q_T} lies within r_* of $G\theta_*$, and local rigidity, with $\mu_T \geq \mu/2$, identifies it with $G\theta_T$. □

Consequences. Let θ_{opt} be the population optimum and $\Delta_T = R_P(\theta_T) - R_P(\theta_{\text{opt}})$.

- **Real-data risk vs the optimum.** The reconstruction student's excess over the optimum is exactly Δ_T for all $n \geq n_0$. It never improves on its teacher, at any sample size. The raw ERM's excess is $\frac{1}{2}\operatorname{tr}(H^+\Sigma)/n\,(1 + o(1))$ in expectation. So

$$\text{reconstructions win} \iff n < N_{\text{eff}} := \frac{\operatorname{tr}(H^+\Sigma)}{2\Delta_T}\,(1 + o(1)).$$

- **Compression.** If the teacher itself is the raw ERM on N samples, then $\Delta_T \approx \frac{1}{2}(\theta_T - \theta_{\text{opt}})^\top H(\theta_T - \theta_{\text{opt}})$ with $\mathbb{E}\Delta_T \approx \operatorname{tr}(H^+\Sigma)/(2N)$, so $N_{\text{eff}} \approx N$. **A student trained on n_0 reconstructions inherits the quality that N raw samples bought the teacher, with n_0 independent of N.** A raw student needs $\approx N$ samples to match it.
- **Imitation of the imperfect teacher (target (b)).** The reconstruction student's imitation error is exactly 0. The raw student's converges to $\mathbb{E}\|\pi_{\text{opt}}(X) - \pi_T(X)\|^2 > 0$: it learns a different function.

This is precisely the trade-off seen in the digits pilot. Imitating an imperfect teacher more closely is *the* mechanism, and beyond N_{eff} raw training wins. It is not an artefact: Theorem 4 shows that, in the identifiable regime, reconstruction training cannot do anything else.

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8. The memorisation threshold: what the strongest reading would need

The intuition proposed was that there should be a sample size at which training fully on raw samples overfits, while training fully on reconstructions does not. Theorems 1–3 prove this for all $n \geq n_0$: raw overfits by $\asymp \operatorname{tr}(H^+\Sigma)/n$, while reconstruction training is exact. A stronger version would ask for an n at which raw training *memorises* (zero training loss, large excess) while *every* exact fit to reconstructions is the teacher. A dimension count rules that out.

Proposition 7. Suppose the student has *local interpolation capacity* at the teacher for n points: the map

$$\Phi_n(\theta, z_1, \dots, z_n) = \big(D_\theta(z_1), \dots, D_\theta(z_n)\big) \in \mathbb{R}^{nd}$$

has rank nd at $(\theta_*, z_1^*, \dots, z_n^*)$, where $Y_i = D_*(z_i^*)$. Then:

1. Every raw sample with X_i close enough to Y_i is interpolated exactly by some student near θ_* (raw training can memorise).
2. The exact fits to $Y_1, \dots, Y_n$ near θ_* contain a C^2 submanifold of dimension $p + nk - nd$ through θ_*. If this exceeds $\dim G\theta_*$, there are exact fits arbitrarily close to θ_* with $R_Q > 0$, so the conclusion of Theorem 1 fails at this n.

Moreover the empirical version of (A4) needs $n(d-k) \geq \dim N = p - \dim G\theta_*$, since each $G(Y_i)$ has rank $\leq d-k$. Local memorisation needs $n \leq p/(d-k)$.

Proof. Φ_n is a submersion at the point, so it is open there; this gives (1). It also makes $\Phi_n^{-1}(Y)$ a C^2 manifold of dimension $p + nk - nd$. Its projection to θ is an injective immersion: z_i is locally determined by θ, and the kernel $\{(0, \zeta) : A_i \zeta_i = 0\}$ is trivial because D is an immersion. □

So the window in which raw training can memorise near the teacher *and* reconstruction training is exact is at most $\big[(p - \dim G\theta_*)/(d-k),\ p/(d-k)\big]$. In the memorisation regime, any claim that the reconstruction student does not overfit concerns the minimiser selected by an implicit bias. That is a separate hypothesis: plausibly the teacher's canon is the "simplest" interpolant, because it is a typical trained network. What does hold for every exact fit, of any architecture, is Prop 2.3: $R_P \leq (\sqrt{R_*} + \sqrt{q_n})^2$. There is no analogous statement for raw fits.

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9. Algorithms and basin access

9.1 Local convergence.

Batch. On the event of Theorem 1, the empirical reconstruction loss satisfies $\hat{R}_n^Q \geq \frac{\mu}{8} \operatorname{dist}^2$ within r_* of the orbit, plus a matching local PL inequality: $\|\nabla \hat{R}_n^Q\|^2 \geq c\mu\, \hat{R}_n^Q$, from the same expansion. Consequences:

- Gradient descent from any initialisation within r_*/C of the orbit converges linearly to an *exact* fit. Gauss–Newton with the normal Gauss–Newton matrix converges quadratically, as in the zero-residual case.
- Raw training converges to the raw ERM, whose excess is $\Theta(1/n)$ by Theorem 2.

Theorem 5 (online, local; sketch of proof). Assume (A1), (A2), (A4) and (A3). Let L, L' be smoothness constants of R_Q and of the per-sample losses near the orbit. Consider constant-step SGD on fresh reconstructions, $\theta_{t+1} = \theta_t - \eta\, g_{\theta_t}(Y_t)$ with $\eta \leq \mu/(4LL')$. For every δ there is $r_0 \asymp \delta\, r_* \mu/\sqrt{LL'}$ such that, if $\operatorname{dist}(\theta_0, G\theta_*) \leq r_0$, then with probability $\geq 1 - \delta$ the iterates stay in the local region and

$$\mathbb{E}\big[R_Q(\theta_t); \text{stay}\big] \leq (1 - \eta\mu/4)^t R_Q(\theta_0).$$

For a stationary teacher this gives

$$\mathbb{E}\big[(R_P(\theta_t) - R_*); \mathrm{stay}\big] \le \frac{8L_P}{\mu}(1 - \eta\mu/4)^t R_Q(\theta_0).$$

Theorem 2 gives $\Omega(1/t)$ for raw SGD with any schedule.

Sketch.

- Morse–Bott gives a local PL inequality $\|\nabla R_Q\|^2 \ge \frac{\mu}{2} R_Q$, and Prop 2.2 gives $\mathbb{E}\|g\|^2 \le 2L' R_Q$. The descent lemma then yields $\mathbb{E}[R_Q(\theta_{t+1}) \mid \mathcal{F}_t] \le (1 - \eta\mu/2 + LL'\eta^2) R_Q(\theta_t)$ while inside the region.
- The stopped process is a supermartingale. The expected path length is $\eta \sum_t \mathbb{E}\|g_t\| \lesssim \frac{\sqrt{L'}}{\mu}\sqrt{R_Q(\theta_0)}$.
- Markov's inequality bounds the probability of exiting.

This is the same-optimiser comparison the stated online argument aimed at, made unconditional on the raw side.

9.2 Basin access.

- *Statistically*, basin access is free: Theorem 1 concerns any exact fit.
- *Verifiably*, any procedure — restarts, annealing, any optimiser — whose output reaches reconstruction training loss $\le \tau_0$ is certified by Cor 3.2. With restarts, the expected number of attempts is $1/p_{\mathrm{succ}}$. Here $p_{\mathrm{succ}} > 0$ for any full-support initialisation, because the exact basin has nonempty interior. But nothing makes it large: in §10.3 it was below $1/40$ at $n = 256$ with our optimiser.
- *Algorithmically, from an arbitrary independent initialisation*, basin access cannot be derived from the hypotheses above. The following shows why, and identifies the condition that matters.

Proposition 8 (symmetry-breaking teachers create reconstruction traps). Let θ' be a nondegenerate global minimiser of R_P whose canon differs from M_* on $\operatorname{supp} Q$, i.e. $R_Q(\theta') > 0$. A typical example is $\theta' = g \cdot \theta_*$ for an isometry g of $\mathbb{R}^d$ preserving P but not M_*, realised by the output layer ($W \mapsto gW$, $c \mapsto gc$). Suppose that on a ball $B(\theta', \rho)$:

- R_P and R_Q are C^2;
- $\nabla^2 R_P \succeq \mu'$ on the normal space of the symmetry orbit through θ';
- $x \mapsto \nabla \ell_\theta(x)$ and $x \mapsto \nabla^2 \ell_\theta(x)$ are κ-Lipschitz on the segments $[\pi_*(x), x]$, $x \in \operatorname{supp} P$. This holds when both clouds are within the reach of M_θ.

If $\kappa\sqrt{2R_*} \le \min(\mu'/2,\ \mu'\rho/8)$, then R_Q has a strict local minimiser θ'' with $|\theta'' - \theta'| \le 4\kappa\sqrt{2R_*}/\mu'$ and $R_Q(\theta'') > 0$. If $\nabla R_Q(\theta') \ne 0$ (generic), then $R_P(\theta'') > R_*$.

Proof. $\nabla R_P - \nabla R_Q = \mathbb{E}[\nabla \ell_\theta(X) - \nabla \ell_\theta(\pi_*(X))]$, so $\|\nabla R_Q(\theta')\| \le \kappa\sqrt{2R_*}$ (using $\nabla R_P(\theta') = 0$). Likewise $\nabla^2 R_Q \succeq \mu' - \kappa\sqrt{2R_*} \ge \mu'/2$ on the normal slice. A strongly convex function on a ball whose gradient at the centre is small has an interior minimiser near the centre. Invariance extends minimality off the slice. □

So:

- Gradient training on reconstructions, started in the basin of an inequivalent optimum, converges to θ'' with *constant* excess real-data risk. Raw training from the same initialisation converges to θ', which is

just as good as the teacher.
- With $|\{\text{inequivalent optima}\}| = k_{\text{opt}}$ and symmetric initialisation, the reconstruction student succeeds roughly with probability $1/k_{\text{opt}}$ and is otherwise worse than the raw student.
- The certificate flags the failure, since $\hat{R}_n^Q(\theta'') \approx R_Q(\theta'') > 0$.

So (B1) — uniqueness of the teacher's optimum modulo symmetries, i.e. no data symmetry that the canon must break — is *necessary* for gradient transmission to be robust to initialisation. It is the same condition that makes the global part of Theorem 1 free.

It is not sufficient. In §10.3 the data have no symmetry at all, yet random-initialised reconstruction training stalls in spurious basins. Prop 2.1 explains why the two problems look alike at coarse scale: $\sqrt{R_P}$ and $\sqrt{R_Q}$ agree up to $\sqrt{R_*}$ everywhere, so the coarse basin structure of the reconstruction landscape is essentially that of the raw one. Algorithmic transmission from an independent initialisation therefore inherits the raw problem's own basin-access difficulty, plus the symmetry traps above. What it adds is a certificate.

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10. Numerical experiments

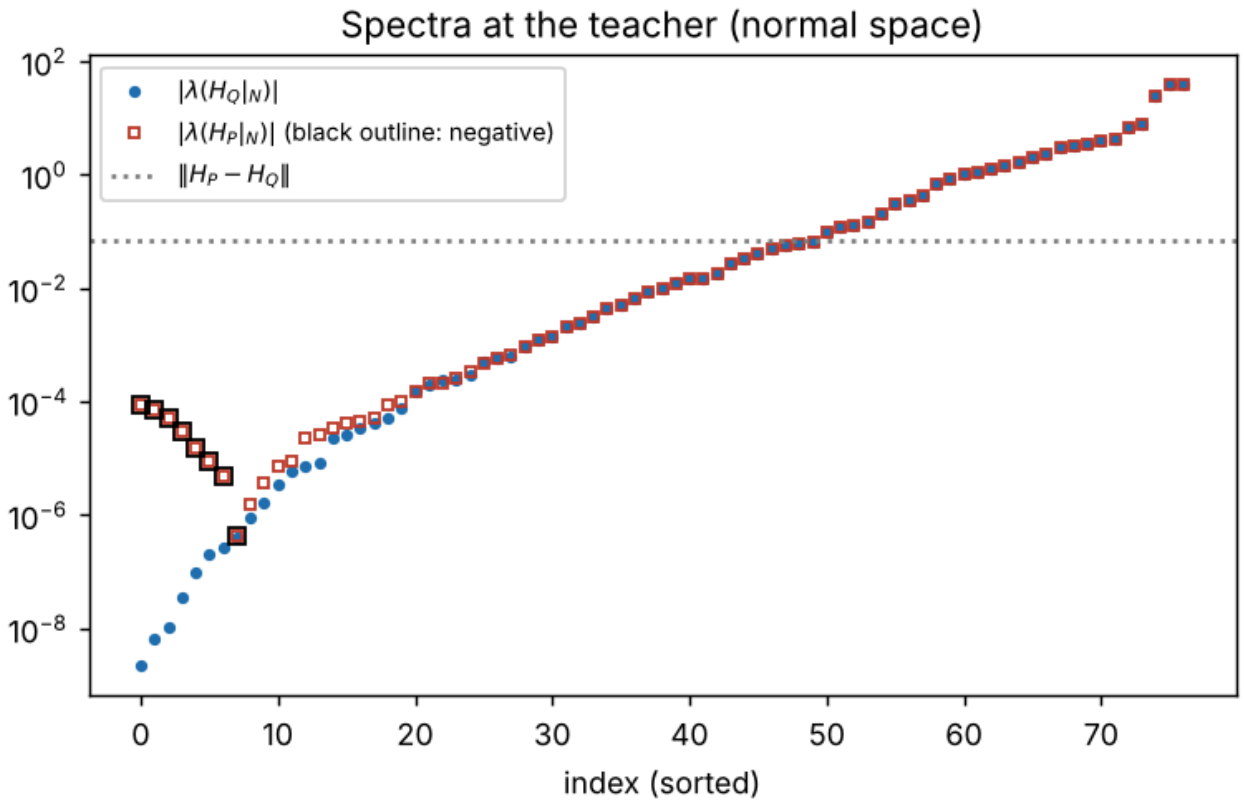
All code is in `research/transmission/code/` (numpy only); see the README there. Every number below is produced by those scripts.

Model.

- Decoder: $D_\theta(t) = W \tanh(A(\cos t, \sin t) + b) + c$, with latent domain the circle (boundaryless), $d = 6$, $m = 8$ hidden units, so $p = 78$.
- Symmetries: latent $O(2)$ and hidden-unit permutations/sign flips; the orbit is 1-dimensional.
- Oracle: 1024-point latent grid plus safeguarded Newton (exact to machine precision away from near-ties). For imitation errors a two-candidate version is used (E1c).
- Truth: $x = \gamma(t) + \sigma\xi$, with γ a random smooth closed curve in $\mathbb{R}^6$ (three harmonics), t von Mises (non-uniform), ξ Gaussian truncated at $3\sqrt{d}$, and $\sigma = 0.05$.
- Teacher: best of 8 restarts, then Levenberg–Marquardt on $N = 40{,}000$ samples ("samples-saturated").
- Evaluation: a fresh test set of $40{,}000$, with common random numbers across all students.
- Training "fully": Adam followed by Levenberg–Marquardt with the normal Gauss–Newton matrix and a global re-projection after each step.

10.1 Diagnostics at the saturated teacher (E4; Fig. 5). Test risk $R_* = 6.12 \times 10^{-3}$, so $\sqrt{2R_*} = 0.111$.

- **Q-rigidity (A4) holds but is badly conditioned.** On the 77-dimensional normal space, $\lambda(H_Q)$ ranges over $[2.2 \times 10^{-9},\ 39.4]$. The orbit direction has eigenvalue 10^{-15}: exactly the latent-rotation symmetry, and nothing else.
- **Curvature transfer (Prop 6.1).** $\|H_P - H_Q\| = 0.071$, about $0.65\sqrt{2R_*}$.
- **The flattest directions.** Nine eigenvalues of $H_P|_N$ are slightly negative (down to -9×10^{-5}), at the resolution of a 20,000-sample estimate. In its flattest directions the teacher is numerically degenerate rather than strictly convex: (B3) holds only up to near-flat directions, as one expects for trained networks.
- **Transfer constant.** The linear-transfer constant $\max_v v^\top H_P v / v^\top H_Q v$ is ≈ 900.
- **Predicted raw constant.** $\frac{1}{2}\operatorname{tr}(H^+\Sigma) = 0.0725$ (with H_Q in place of H_P: 0.0726).

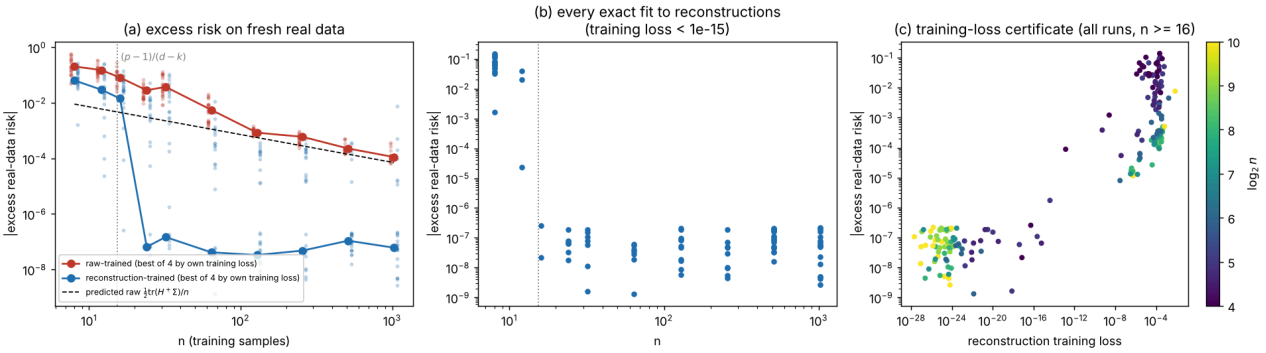


E4. Spectra of H_Q and H_P on the normal space at the saturated teacher. They agree down to $\approx \|H_P - H_Q\|$. Below that, H_Q stays positive (rigidity) while H_P is flat or slightly negative.

10.2 Students started in the teacher's basin (E1; Fig. 1). Initialisation is teacher $+\mathcal{N}(0, 0.05^2 I)$, with 6 data seeds $\times$ 4 initialisations; each student keeps its best run by *its own* training loss. This tests Theorems 1–3 and Prop 7. It is not an independent initialisation.

n	exact fits among rec. runs	max $\lvert$excess$\rvert$ over exact fits	rec. excess (median, best-of-4)	raw excess (median, best-of-4)	raw / predicted	raw test − train gap
8	15/24	1.5×10^{-1}	6.5×10^{-2}	2.1×10^{-1}	22.7	2.1×10^{-1}
12	3/24	4.0×10^{-2}	3.0×10^{-2}	1.5×10^{-1}	25.4	1.6×10^{-1}
16	2/24	2.6×10^{-7}	1.4×10^{-2}	8.2×10^{-2}	18.1	8.8×10^{-2}
24	7/24	1.8×10^{-7}	6.5×10^{-8}	2.9×10^{-2}	9.5	3.2×10^{-2}
32	7/24	1.9×10^{-7}	1.5×10^{-7}	3.8×10^{-2}	16.7	4.1×10^{-2}
64	7/24	6.2×10^{-8}	4.2×10^{-8}	5.6×10^{-3}	5.0	8.0×10^{-3}
128	11/24	2.1×10^{-7}	3.3×10^{-8}	8.5×10^{-4}	1.5	2.0×10^{-3}
256	9/24	1.9×10^{-7}	4.8×10^{-8}	6.0×10^{-4}	2.1	1.0×10^{-3}
512	18/24	2.1×10^{-7}	1.1×10^{-7}	2.3×10^{-4}	1.6	6.6×10^{-4}
1024	17/24	2.2×10^{-7}	6.1×10^{-8}	1.1×10^{-4}	1.6	2.7×10^{-4}

"Exact fit" means reconstruction training loss $< 10^{-15}$; observed values are 10^{-28}–10^{-17}. "Predicted" is $\frac{1}{2}\,\mathrm{tr}(H^+\Sigma)/n$ with the constant measured at the teacher.



E1. (a) Excess real-data risk; faint points are individual runs. (b) Every exact fit to reconstructions: identifying from $n \geq 16$, not below. (c) Training loss versus excess for all reconstruction runs with $n \geq 16$.

Target (b), measured carefully (E1c). The training runs use a single-candidate oracle. That oracle can mis-assign test points that are nearly equidistant from two branches of the curve: this costs $\leq 3 \times 10^{-7}$ in risk, but a full branch separation in the imitation metric. So the E1 and E3 students were re-trained with the same seeds and saved, then evaluated with a two-candidate oracle (`exp_imitation.py`). For every exact fit:

students of	n	exact fits	max $\lvert$excess$\rvert$	max imitation error $\mathbb{E}\lVert\pi_{\hat{\theta}}(X) - \pi_{\theta_T}(X)\rVert^2$	raw students' imitation error (median)
saturated teacher	32	3/12	3.5×10^{-9}	1.5×10^{-6}	7.6×10^{-2}
saturated teacher	128	6/12	1.8×10^{-9}	1.5×10^{-6}	2.9×10^{-3}
saturated teacher	512	9/12	5.1×10^{-9}	2.3×10^{-6}	1.5×10^{-3}
$N = 1024$ teacher	128	3/6	9.2×10^{-9}	1.1×10^{-6}	—
$N = 1024$ teacher	512	5/6	9.2×10^{-9}	2.1×10^{-6}	—

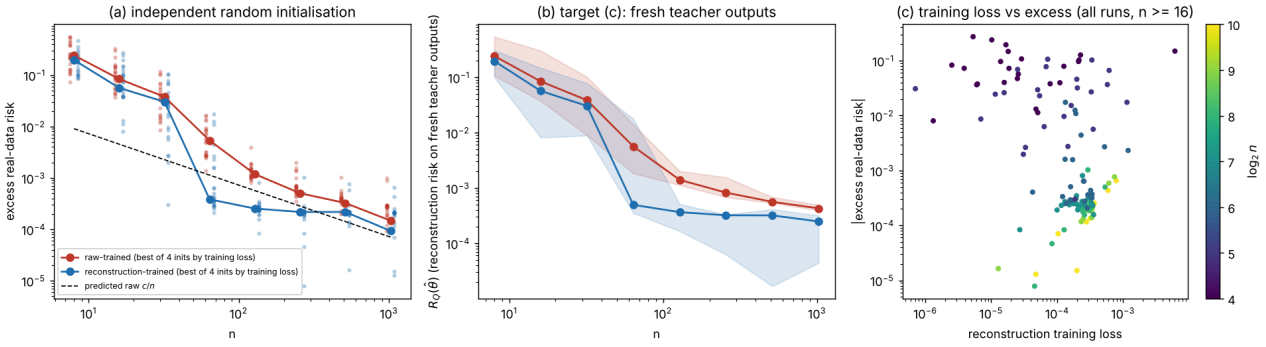
The residual imitation error of exact fits comes from 1–4 of the 40,000 test points: genuine near-ties at the medial axis, a P-null set in theory. On the reconstructions themselves $R_Q \leq 4 \times 10^{-13}$, and $\leq 10^{-21}$ for training losses below 10^{-23}.

- **Every exact fit is exact transmission once $n \geq 16$.** Every reconstruction student whose training loss reaches numerical zero ($< 10^{-15}$) has $\lvert$excess real-data risk$\rvert \leq 2.6 \times 10^{-7}$. That is the resolution of the evaluation, set by oracle tie-breaking for test points near the medial axis. Exact imitation and R_Q behave the same way.
- **Below the threshold, exact fits are not identifying.** At $n = 8$ and 12 exact fits are abundant but have excess up to 0.15. The switch happens between $n = 12$ and $n = 16$, exactly where the dimension count of Prop 7 puts it: $(p - \dim G\theta_*)/(d - k) = 77/5 = 15.4$.
- **Raw students overfit at the predicted rate.** For $n \geq 128$ their excess is within a factor 1.5–2.1 of the predicted $\frac{1}{2}\,\mathrm{tr}(H^+\Sigma)/n$, and their test–train gap is about twice the excess. The best-of-4 reconstruction student is exact in the median seed for every $n \geq 24$. The separation is 3–6 orders of magnitude.
- **Optimisation failures are visible.** Some runs from this initialisation still stall at a local minimum of the empirical reconstruction loss. All of them have training loss $\geq 10^{-8}$ (Fig. 1c), so the certificate

separates success from failure without overlap.

10.3 Independent random initialisation (E2; Fig. 2). Same comparison, but from the standard random initialiser (4 per seed, shared by both students), with Adam then LM.

n	raw excess (median, best-of-4)	rec. excess (median, best-of-4)	raw / rec.	rec. runs with training loss $<$ 10^{-5}
8	2.4×10^{-1}	2.0×10^{-1}	1.2	24/24 (exact, but non-identifying)
16	8.4×10^{-2}	5.6×10^{-2}	1.5	6/24
32	3.8×10^{-2}	3.0×10^{-2}	1.3	2/24
64	5.3×10^{-3}	3.8×10^{-4}	14	0/24
128	1.2×10^{-3}	2.5×10^{-4}	4.7	0/24
256	5.0×10^{-4}	2.2×10^{-4}	2.3	0/24
512	3.3×10^{-4}	2.2×10^{-4}	1.5	0/24
1024	1.5×10^{-4}	9.3×10^{-5}	1.6	0/24

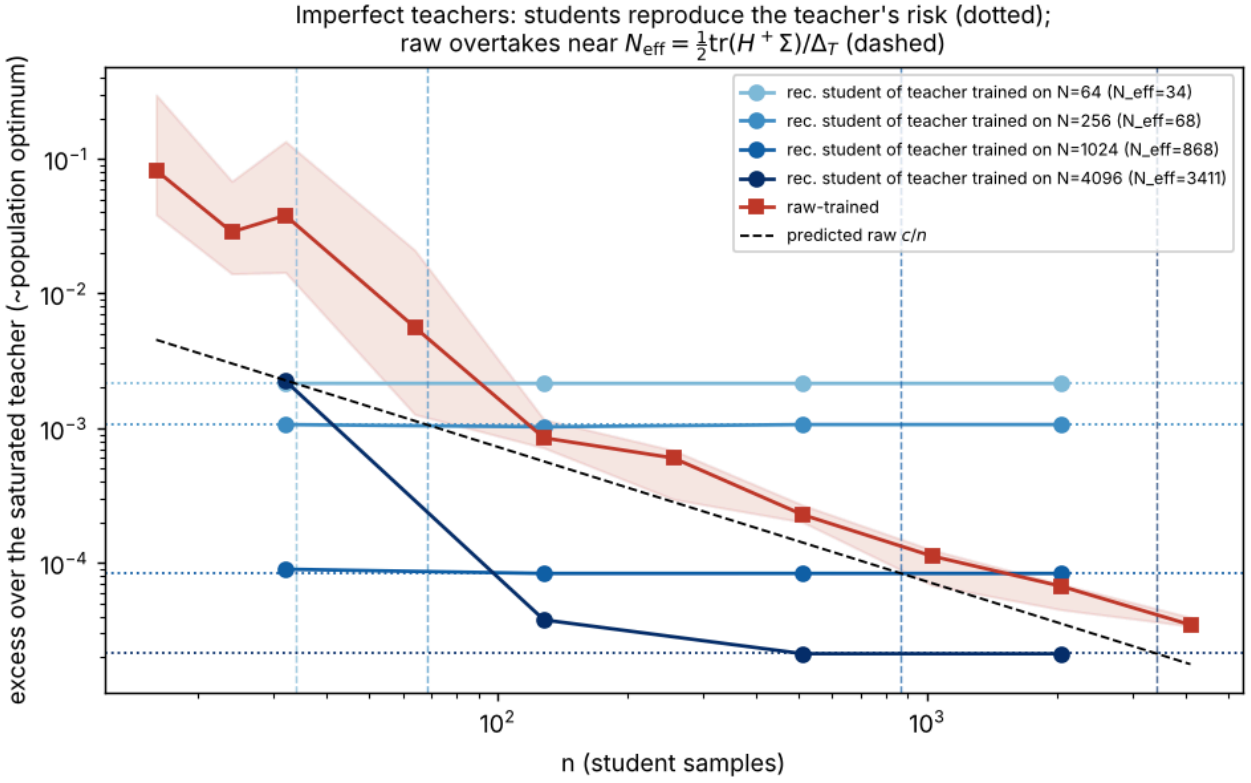


E2. (a) Excess real-data risk from independent random initialisations; faint points are individual runs. (b) Reconstruction risk on fresh teacher outputs. (c) Training loss versus excess for reconstruction students.

- **No exact fits from random inits.** For $n \geq 16$, *no* reconstruction student reached an exact fit in this budget (0 of 168 runs). Most stall in a spurious basin with training loss $\approx 3 \times 10^{-4}$ and excess $\approx 2 \times 10^{-4}$, and the training loss flags all of them.
- **Still ahead of raw, but by a shrinking factor.** They beat the raw students at every $n \leq 1024$: by $14\times$ at $n = 64$, $4.7\times$ at $n = 128$, and about $1.6\times$ at $n = 1024$. Raw students also land in somewhat worse basins: their excess is $\approx 2\times$ the in-basin prediction.
- **This is a finite window.** The reconstruction students' floor is a property of their optimisation, not of n. Extrapolating the raw curve, raw training overtakes them near $n \approx 2000$. It is the algorithmic analogue of the imperfect-teacher window, with the student's basin error in place of the teacher's.
- **Longer runs** (E2b, 40 restarts at $n = 256$ with a longer schedule): none of the 40 reconstruction students reached an exact fit. Their median training loss was 3.5×10^{-4} and median excess 2.5×10^{-4}. Selecting by own training loss over all 40 restarts gives a reconstruction student with excess 1.4×10^{-5} (training loss 4.5×10^{-5}). The same rule picks a raw student with excess 7.7×10^{-4}, *worse* than the raw median (6.5×10^{-4}): the lowest raw training loss belongs to the most overfit run. That is the certificate asymmetry, seen directly.

10.4 Imperfect teachers (E3; Fig. 3). Teachers are raw ERMs on $N \in \{64, 256, 1024, 4096\}$ samples, started at the saturated teacher. Their reconstruction students are started near their own teacher (3 initialisations per seed); raw students are those of E1 plus extra runs at $n = 2048, 4096$.

teacher's N	teacher's excess Δ_T	$N_{\text{eff}} = \frac{1}{2}\operatorname{tr}(H^{+}\Sigma)/\Delta_T$	exact fits among rec. runs	rec. excess (median, best-of-3)	max $\lvert$rec. excess $-\Delta_T\rvert$ over exact fits	first n at which raw median $< \Delta_T$
64	2.15×10^{-3}	34	7/24	2.15×10^{-3}	4.9×10^{-8}	128
256	1.06×10^{-3}	68	14/24	1.06×10^{-3}	8.7×10^{-7}	128
1024	8.36×10^{-5}	868	14/24	8.35×10^{-5}	1.1×10^{-7}	2048
4096	2.13×10^{-5}	3411	8/24	2.13×10^{-5}	1.8×10^{-7}	none up to 4096



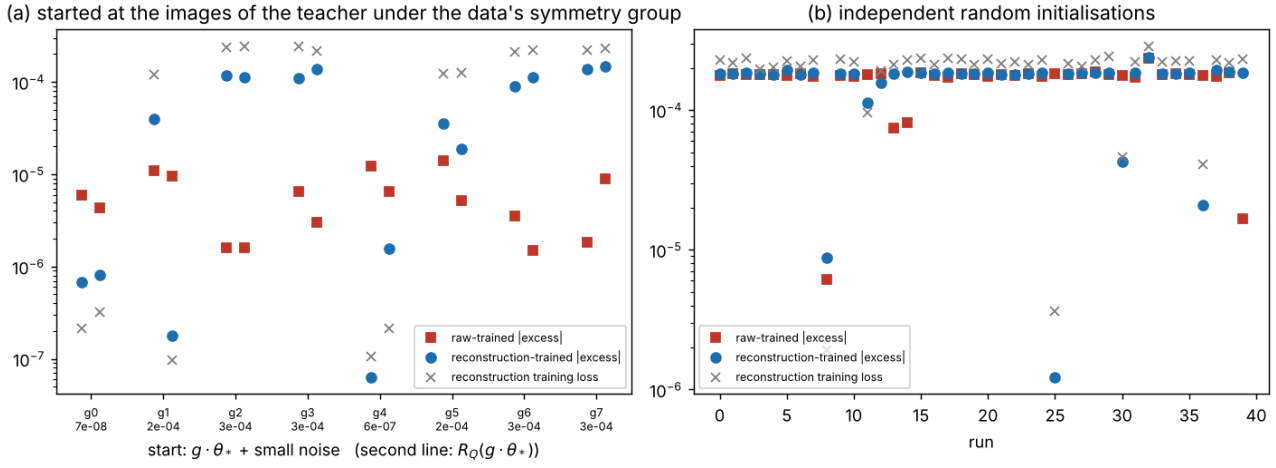
E3. Reconstruction students of imperfect teachers (blue) sit exactly at their teacher's excess (dotted). Raw students (red) overtake shortly after N_{eff} (dashed).

- **The teacher's risk is transmitted exactly (Theorem 4).** Every exact fit reproduces its teacher's real-data excess to within 9×10^{-7}, from $n = 32$ reconstructions on, for teachers trained on 64 to 4096 samples.
- **Imitation is exact too.** E1c shows the imitation error of the imperfect teacher's map is $\sim 10^{-6}$.
- **Raw students overtake on schedule.** They cross each teacher's level at $n \approx (1\text{–}2)\, N_{\text{eff}}$. The factor reflects raw excess running at about $1.6\times$ the asymptotic prediction.

- **The window is set by the teacher's actual excess, not by N.** The $N = 256$ teacher happened to be unusually poor ($\Delta_T \approx 3.8\times$ its expected value), and its window closed correspondingly early. With an average teacher, $N_{\text{eff}} \approx N$.
- **Compression.** For the $N = 4096$ teacher, 32–512 reconstructions give the student the teacher's quality. Raw students need more than 4096 samples to match it.

10.5 Symmetry-breaking traps (E5; Fig. 4).

- *Setup.* P is a noisy superellipse $|x|^4 + |y|^4 = 1$ in $\mathbb{R}^2$, which has dihedral symmetry D_4, with $\sigma = 0.04$. The decoder has $m = 3$ hidden units, so its closed curve cannot be D_4-symmetric. Each of the 8 images $g \cdot \theta_*$ ($W \mapsto gW$, $c \mapsto gc$) has the teacher's raw risk to within $\pm 4 \times 10^{-6}$ (test-set noise).
- *Which images are inequivalent.* Six of them are inequivalent canons: $R_Q(g\theta_*) = 1.6 \times 10^{-4}$ (two of them) and 3.3×10^{-4} (four). The remaining reflection is numerically a symmetry of the teacher's curve ($R_Q = 6 \times 10^{-7}$).
- *Started at the four strongly inequivalent optima (8 runs).* All 8 raw students converge to equally good optima (excess $\leq 1.2 \times 10^{-5}$, the $n = 2000$ estimation noise). All 8 reconstruction students are **trapped**: training loss stays at 2.2–2.4×10^{-4} and excess real-data risk at 0.9–1.5×10^{-4}. They do worse than the raw students by roughly an order of magnitude, exactly as Prop 8 predicts.
- *From the milder ones.* Three of four are trapped.
- *From the teacher's own basin and its near-symmetry.* All four reconstruction students reach the teacher, at the oracle's precision floor in this example.
- *The certificate separates the two groups cleanly.* Trapped runs keep training loss $\geq 10^{-4}$; successful ones end at $\leq 3 \times 10^{-7}$.
- *From random initialisations.* With so few units the raw landscape has its own dominant spurious basin (excess $\approx 1.8 \times 10^{-4}$), into which both kinds of student fall in about 80% of runs.



E5. (a) Started at the images $g \cdot \theta_*$ of the teacher under the data's symmetry group; the second label line is $R_Q(g \cdot \theta_*)$. Raw students (red) converge to equally good optima; reconstruction students (blue) are trapped wherever $R_Q(g \cdot \theta_*)$ is large, and their training loss (crosses) shows it. (b) Independent random initialisations.

10.6 Wide decoder without a capacity bound (E4b). A wider decoder ($m = 10, p = 96$) has no capacity bound and its teacher has $\|\theta\| = 10$.

- *Started near the teacher.* Students converge quadratically to exact fits (training loss $\sim 10^{-28}$ within 25–32 LM iterations) from perturbations up to 0.1 per coordinate.

- *From random initialisations.* The training loss plateaus near 10^{-6}. Meanwhile $\|\theta\|$ grows from about 9 to 37–59, the second eigenvalue of the normal Gauss–Newton matrix collapses from $\sim 10^{-9}$ to $\sim 5 \times 10^{-12}$, and the largest grows to $\sim 10^{2.5}$–10^{3}.
- *Reading.* These are near-zero valleys at infinity. In an unbounded parameter space, the global margin γ of Theorem 1 is (numerically) zero. Compactness, i.e. a capacity class shared by teacher and student, is doing real work in the theorem.

10.7 What the numerics add.

- The statistical content of Theorems 1–3 shows up sharply wherever training actually finds an exact fit: exact transmission, at numerical precision, from the predicted threshold $n \approx (p-1)/(d-k)$ onward.
- Raw students follow the predicted $\Theta(1/n)$ law and overfit.
- The binding constraint in practice is algorithmic basin access, which the theory isolates but does not supply. From independent initialisations the advantage was real but bounded by the student's optimisation floor.
- Symmetry breaking is a concrete, reproducible mechanism by which basin access fails for reconstruction training specifically. In all of these cases the reconstruction training loss told the student which situation it was in.

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11. What remains open

1. **(A3)/(B1) for a concrete class.** For example: a range-identifiability theorem for two-layer tanh decoders with generic weights on a boundaryless latent space (the analogue of Sussmann's theorem for input–output maps). Lemma 6.2 reduces (A3) to unique optimality. What is missing is a structural theorem saying when unique optimality modulo symmetries holds.
2. **Quantitative n_0.** Theorem 1 is exponential in n, but its constants include the global margin γ and covering numbers of $\operatorname{supp} Q$ at scale $\sqrt{\gamma}$. Lemma 6.2 bounds γ through the raw margin. Polynomial control of n_0 in terms of p, $d-k$ and μ needs landscape information.
3. **Algorithmic basin access.** Prop 8 shows it fails under symmetry breaking. Under (B1) it is plausible that the two-sided transfer (Prop 2.1), which makes $\sqrt{R_P}$ and $\sqrt{R_Q}$ uniformly $\sqrt{R_*}$-close, lets gradient training on reconstructions inherit whatever basin access raw training has. A natural statement: the gradient fields are uniformly $O(\sqrt{R_*})$-close wherever projections are stable, so outside $O(\sqrt{R_*})$-neighbourhoods of raw critical points the two flows decrease each other's objectives. Turning this into a theorem needs control near raw saddles.
4. **Memorisation regime.** Prop 7 says any separation there is an implicit-bias statement. A model in which it can be proved (e.g. minimum-norm or NTK-type selection among exact fits) would settle the strongest reading of the batch intuition.
5. **Real data.** A saturated teacher cannot be obtained from a finite dataset such as sklearn digits. The relevant prediction (Thm 4) is the crossover at $n \approx N_{\text{eff}}$, where the teacher's effective sample size is measured from its excess risk. Testing this on real data needs a teacher trained on a much larger pool than the students.

Appendix A. Hessian of the profiled loss

Let $f(\theta, z) = \tfrac{1}{2}\|x - D_\theta(z)\|^2$, and suppose $z_\theta(x)$ is a nondegenerate interior minimiser. Then $\ell_\theta(x) = f(\theta, z_\theta(x))$, the envelope theorem gives $\nabla_\theta \ell = \partial_\theta f = -J^\top r$, and

$$\nabla^2_\theta \ell = f_{\theta\theta} - f_{\theta z} f_{zz}^{-1} f_{z\theta},$$

$$f_{\theta\theta} = J^\top J - \langle r, D_{\theta\theta}\rangle, \qquad f_{\theta z} = J^\top A - \langle r, D_{\theta z}\rangle, \qquad f_{zz} = A^\top A - \langle r, D_{zz}\rangle.$$

At a zero residual ($x = y \in M_\theta$) this is $J^\top J - J^\top A (A^\top A)^{-1} A^\top J = J^\top P^\perp J$.

The same Schur complement gives the reduced normal equations used by the Levenberg–Marquardt trainer in the code: eliminating the latent increments from the joint Gauss–Newton system leaves $\big(\sum_i J_i^\top P_i^\perp J_i + \lambda \operatorname{diag}\big)\Delta\theta = \sum_i J_i^\top r_i$.

Appendix B. Reproducing the experiments

See `research/transmission/code/README.md`.